Chapter 22 — Revision: Cheat Sheets & Glossary

> **What this chapter is.** The whole handbook on a few pages: the answer framework (Ch 1); OOP, relationships and UML (Ch 2–3); SOLID, other principles and clean code (Ch 4–6); all 24 smells (Ch 7); all 23 GoF patterns + extras and a "which pattern?" table (Ch 8–11); the concurrency toolbox (Ch 12–13); a map of every case study (Ch 14–21); machine-coding and edge-case checklists; and a glossary. Nothing here is new and there is no code — each row tells you where the full explanation lives.

How to read the references. "Ch 12 · Deadlock" means the *Deadlock* topic in Chapter 12; "Ch 16 · Car Rental" means the Car Rental case study in Chapter 16. When a row or heading already names the topic, only the chapter is given ("Ch 2"). Numbers such as "22.7" refer to sections of this chapter.

22.1 How to revise

In 15 minutes (the morning of the interview)

Minutes	Read	Why
0–3	22.2 (the framework)	You will use it in the first minute of the interview
3–6	22.4 (SOLID one-pager) and the relationship table in 22.3	The two things asked about most often by name
6–10	"Which pattern?" table in 22.7	Patterns should come from a signal, not from memory
10–13	Concurrency cheat sheet and deadlock checklist in 22.8	Every booking/inventory problem has a race
13–15	The row(s) of 22.9 for the problems you expect	One sentence per problem: entities, pattern, the atomic step

The night before (about 2 hours)

1. **Read 22.2–22.8 slowly** (45 min), saying *one handbook example* per row; if you can't, open the referenced topic.
2. **Three unpractised case studies** from 22.9 (45 min): requirements, classes, class diagram and the one hard method, without looking; then compare with the chapter's TL;DR.
3. **Scan the glossary** (22.11); read up only the terms you can't explain in one sentence (20 min).
4. **Stop.** Sleep matters more than one more pattern.

Common mistakes when revising

- Re-reading chapters instead of **recalling** first and then checking. Recall is what the interview tests.
- Memorising all 23 patterns equally. Six come up in almost every round (22.7): Strategy, Factory, Observer, State, Singleton, Builder.
- Skipping the "when *not* to use it" column. Interviewers ask "would you always do that?" (Ch 4 · SOLID in interviews).

22.2 The LLD interview answer framework (one page)

Source: Chapter 1 (how the interview runs, the answer framework, requirements, classes, SOLID check, patterns, interview code, communication tips, common mistakes).

TL;DR

Clarify → requirements → use cases → entities → relationships → class diagram → key flows → patterns/SOLID → code → edge cases/concurrency → extensions. Say each step out loud as you enter it.

The eleven steps

#	Step	Ask yourself	Output
1	Clarify	Who uses it? Single site or many?	Shared understanding
2	Requirements	Functional? Non-functional (thread-safe, extensible, in-memory)? **Out of scope?**	8–12 numbered lines + out-of-scope list
3	Use cases	What can each actor do?	Actor → use-case list
4	Entities	Which nouns hold state or behaviour? Which are just values?	Classes, one sentence each
5	Relationships	Uses / knows / has / owns / is-a? How many?	Typed lines with multiplicity
6	Class diagram	Fields, key methods, interfaces, enums	UML class diagram
7	Key flows	Steps of the 2–3 main use cases; lifecycle of the core object	Sequence / activity / state diagram
8	Patterns & SOLID	A rule that varies? A lifecycle? Many listeners? Two reasons to change?	1–2 named patterns + one rejected
9	Code	Interfaces, enums, models, services, driver	Running code for the main flows
10	Edge cases & concurrency	Bad input? Full? Two users, one seat?	Validation, exceptions, locks
11	Extensions	"What if X is added?"	Short answers, maybe one class

Time budgets

45-minute OOD round		**90-minute machine-coding round**	
Clarify requirements	5–8 min	Read, requirements, assumptions	5–10 min
Actors and use cases	3–5 min	Model: entities, enums, service interfaces	10–15 min
Find the classes	8–10 min	Core flow end to end	30–40 min
Connect the classes	8–10 min	Remaining features, validation, edge cases	15–20 min
Write the code	12–15 min	Polish, one extension point, demo	5–10 min
		Buffer	5 min (never zero)

Rule of thumb: roughly **a quarter of the hour goes before you name a class**, and in machine coding you **get something running by the halfway mark** (Ch 1 · The interview).

Finding classes (Ch 1 · Finding the classes)

- **Nouns → candidate classes; verbs → methods.** A verb with no class to live on means a class is missing.
- Cut candidates with five tests: is it a **value**? the **same thing** under another name? does it have **changing state**? does anything **call** it? does it differ in **behaviour or only data**?
- **Different data → field or enum. Different behaviour → class.**
- One sentence per class, **no "and"**. Put each method where its data is (Information Expert).
- Beginners most often forget the **record-keeping class** (ticket, loan, booking) and the **rule class** (rate, strategy).

What the interviewer judges (Ch 1 · The interview)

1. Did you **bound** the problem? 2. Does each class have **one job** you can say in a sentence? 3. Are the **relationships** right? 4. Does the **code match the diagram**? 5. Can you explain a **trade-off** (the choice and the rejected option)?

Machine coding adds: does it **run**, is it **modular**, **readable**, does it **handle bad input**, and (senior roles) is shared state **thread-safe**.

Code habits interviewers notice (Ch 1 · Interview code)

Habit	Instead of
Pass time in (`long nowMinute` , `Instant now` , injected `Clock`)	Reading the clock inside logic
Money as `BigDecimal` from strings, or whole cents in a `long`	`double` (0.1 + 0.2 ≠ 0.3)
`IllegalArgumentException` / `IllegalStateException` / a custom exception	Silent failure or error codes
`Optional` when "not found" is normal	Returning `null`
`private final` fields	Public mutable fields
Domain names (`ParkingTicket`)	`Manager` , `Helper` , `Data2`

Skeleton: **Models → Repositories → Strategies → Services → Driver** (Ch 1 · Interview code).

In the interview

30-second answer (from Ch 1 · Common mistakes): "I clarify and write 8 to 12 requirements with an out-of-scope list, list actors and use cases, pick classes from the nouns and give each one responsibility, connect them and draw a class diagram, sketch the key flow, run a quick SOLID check and pick one or two patterns where the signal is clear, then code the two or three core flows, handle edge cases and concurrency, and finish with extensions and trade-offs."

Common mistakes (Ch 1)

Mistake	Fix
Coding in minute two	8–13 minutes on requirements and actors first
No out-of-scope list	Write it down and agree it
One subclass per value (`CompactSpot` , `LargeSpot`)	Different data → field/enum
God class (`ParkingLotManager`)	One sentence per class; split on "and"
Inheritance for reuse ("Car extends Engine")	Composition
Pattern stuffing	One or two patterns, each with its signal
All twelve classes at equal depth	Two or three flows fully
Ignoring two users on one resource	Lock or atomic operation on the shared state
Non-running code in machine coding	Thin end-to-end path running by halfway

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22.3 OOP & relationships cheat sheet

Source: Chapter 2 (OOP) and Chapter 3 (relationships and UML).

The four pillars (Ch 2)

Pillar	One line	Java tools	Smell when missing
Encapsulation (Ch 2)	Data is private; only the owning class's methods change it, and they enforce the rules	`private` fields, no setters, validation in constructors	Public fields; setters that break invariants
Abstraction (Ch 2)	Expose only what callers need; hide how	Interfaces, abstract classes, small public API	Callers reach into internals
Inheritance (Ch 2)	Child reuses and specialises parent — **only when "child is a parent" is truly correct**	`extends`, `super`, `@Override`	Subclass refuses or throws on parent methods
Polymorphism (Ch 2)	One call, different behaviour per object; removes type switches	Overriding + dynamic dispatch	`switch` / `instanceof` chains on type

Encapsulation vs abstraction: hiding *data* behind rule-enforcing methods vs hiding *complexity* behind a minimal set of operations (Ch 2 self-check).

Other OOP facts worth one line each

Topic	Remember	Where
Overloading vs overriding	Compile time, by declared parameter types vs run time, by the actual object. Fields and `static` methods are not polymorphic (statics are hidden, not overridden)	Ch 2 · Compile-time vs runtime polymorphism
`this`, `static`, `final`	Current object; one per class; cannot be reassigned/overridden/extended	Ch 2 · Classes & keywords
Interfaces	Contracts; can have `default`, `static`, `private` methods; a functional interface has one abstract method (lambdas)	Ch 2
Enums	Fixed sets with fields and behaviour ("which spot types fit a truck" lives on `VehicleType`)	Ch 2 · Enums with fields and behaviour
Composition over inheritance	Avoids fragile base classes; allows run-time swapping	Ch 2, Ch 5 · Composition over inheritance
`equals` / `hashCode`	Equal objects must have equal hash codes; override both or hash collections break	Ch 2 · Object identity vs equality
Immutability & records	Final class, private final fields, no setters, defensive copies. Records are only **shallowly** immutable — copy collections in the compact constructor	Ch 2 · Immutability & records
Access modifiers	Default to `private`; package-private = same package only	Ch 2 · Access modifiers and packages
Generics	Type-safe reusable infrastructure; **PECS:** producer `extends`, consumer `super`	Ch 2 · Generics basics for LLD
Coupling & cohesion	Low coupling (between classes), high cohesion (inside a class)	Ch 2 · Coupling and cohesion
OOA → OOD → OOP	What the problem is → how, with objects → code. Connections that carry data become objects	Ch 2 · Object-oriented analysis and design

Abstract class vs interface (Ch 2 · Abstract classes vs interfaces)

	Abstract class	Interface
Instantiable?	No	No
Instance fields (state)	Yes	No (only `public static final` constants)
Constructors	Yes	No
Method bodies	Any	`default`, `static`, `private` only
How many per class	**One** (`extends`)	**Many** (`implements`)
Relationship expressed	"is a" (shared identity and state)	"can do" (capability / role)
Typical use	Shared state + partial implementation for a close family (Template Method)	Contracts, roles, strategies, test seams

Rule of thumb: start with an interface; add an abstract class underneath when implementations share real state or a fixed algorithm (JDK: `List` + `AbstractList` + `ArrayList`).

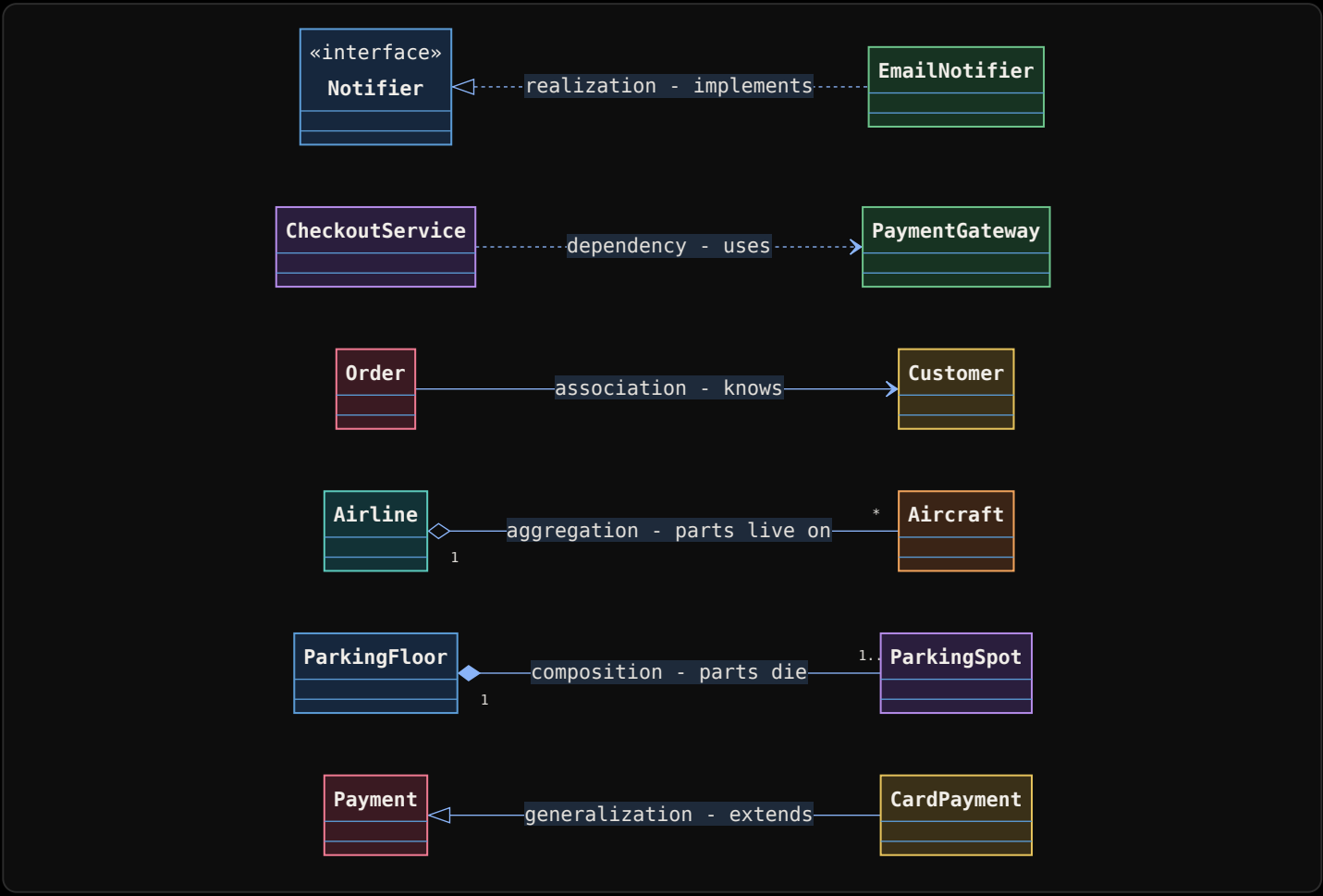
The six relationships (Ch 3)

Weakest to strongest coupling: **dependency, association, aggregation, composition, realization, generalization** (Ch 3 · self-check).

Relationship	Phrase	UML line	Java shape	Lifetime	Example
Dependency (Ch 3)	A *uses* B	dashed, open arrow → B	B as parameter / local / return / static call	none	`CheckoutService` uses `PaymentGateway`
Association (Ch 3)	A *knows* B	solid line (arrow for one-way)	field of type B, created elsewhere	independent	`Order` → `Customer`
Aggregation (Ch 3)	A *has* B, parts live on	solid, hollow ◇ at A (the whole)	field/collection, parts passed in	B outlives A	`Airline` ◇— `Aircraft`
Composition (Ch 3)	A *owns* B	solid, filled ◆ at A (the whole)	A creates B, hides it, B not shared	B dies with A	`ParkingFloor` ◆— `ParkingSpot`
Realization (Ch 3)	A *implements* B	dashed, hollow △ at B	`implements`	—	`EmailNotifier` ⇢ `Notifier`
Generalization (Ch 3)	A *is a* B	solid, hollow △ at B	`extends`	—	`CardPayment` → `Payment`

Choosing, in order (Ch 3 · Choosing the relationship): interface A promises to fulfil → realization. Substitutable with no refused methods → generalization. Not kept in a field → dependency. Part that dies with the whole → composition. Part that outlives it → aggregation. Otherwise → association.

UML arrow legend (Mermaid)



Mermaid syntax: `A ..> B` dependency · `A --> B` association (navigable A→B; `A -- B` unspecified) · `A o-- B` aggregation and `A *-- B` composition (A is the whole) · `B <|.. A` realization · `B <|-- A` generalization · `A "1" *-- "1..*" B` adds multiplicities.

Multiplicity → code (Ch 3 · TL;DR): `1` required field; `0..1` `Optional` /nullable; `1..*` non-empty collection checked in the constructor; exact numbers checked in constructors. Visibility: `+` public, `-` private, `#` protected, `~` package.

UML diagrams in the interview (Ch 3 · UML diagrams)

Diagram	Use it for	Budget in 45 min
Use case (Ch 3 · Use-case diagrams)	Scope: actors + what each does. «include» = always; «extend» = sometimes	Usually spoken, 3–5 min
Class (Ch 3 · Class diagrams)	**The answer.** 10–15 boxes, every line typed, with multiplicity	Always drawn, 10–15 min
Sequence (Ch 3 · Sequence diagrams)	Call order between objects (3–4 objects, 6–8 messages)	One behaviour diagram only
Activity (Ch 3 · Activity diagrams)	A flow that branches; fork/join for parallel steps	"
State machine (Ch 3 · State machine diagrams)	One object's lifecycle; transition label `event [guard] / action`	"

UML has 14 diagram types: 7 structural, 7 behavioural (Ch 3 · What UML is).

Common mistakes

- Diamond at the wrong end (it goes at the **whole**); triangle at the wrong end (it goes at the **parent/interface**) (Ch 3 · Choosing the relationship).

- Lines without type or multiplicity (Ch 3 · Choosing the relationship).
- Inheritance used to share data; subclasses that only pass a constant to `super` → one class with an enum field (Ch 3 · self-check).

22.4 SOLID one-pager

Source: Chapter 4 (the five principles, SOLID and patterns, SOLID in the interview, over-engineering); smells from Chapter 7 (smell lookup table).

TL;DR

SOLID = five names for the ways code becomes painful to change. **Use them to diagnose, not as a checklist**, apply a fix when a real change asks for it, and always say what the fix costs.

Principle	One-line rule	Smell that signals a violation	Fix	Patterns that help
S — Single Responsibility (Ch 4 · Single Responsibility Principle)	A class should have one reason to change (one actor asks for its changes)	The responsibility sentence needs an "and"; *Divergent Change*, *Large Class*	Split by actor (Extract Class); each new class needs a name that stands alone	Command, Proxy, Decorator (each concern in its own class)
O — Open/Closed (Ch 4 · Open/Closed Principle)	Add new behaviour by adding code, not by editing working code	A `switch` / `if` chain over a type; *Repeated Switches*, *Shotgun Surgery*	Put the part that *has* varied behind an interface; add variants as classes (Replace Conditional with Polymorphism)	Strategy, Template Method, Decorator, Factory Method, Observer, Command, Visitor
L — Liskov Substitution (Ch 4 · Liskov Substitution Principle)	A subtype must be usable anywhere its parent is expected	Subclass throws, does nothing, or refuses a parent method; *Refused Bequest*	Better parent, a narrower interface, or composition (Replace Subclass with Delegate)	Composite, Null Object
I — Interface Segregation (Ch 4 · Interface Segregation Principle)	No client should depend on methods it does not use	Empty method bodies or `UnsupportedOperationException` in implementers; fat interfaces	Split into **role interfaces**, one per kind of caller	Adapter
D — Dependency Inversion (Ch 4 · Dependency Inversion Principle)	Depend on abstractions, not concrete details; the policy owns the abstraction	A high-level class does `new` of (or names) a low-level one; cannot test without the real DB/provider	Inject an interface owned by the high-level class; `main` wires the details	Strategy, Factory Method / Abstract Factory, Adapter, Observer

LSP in detail (Ch 4 · Liskov Substitution Principle, TL;DR): a subclass must accept at least as much, promise at least as much, keep invariants and history, and throw nothing new. Square/Rectangle is worse than `Penguin.fly()` throwing because it fails **silently** (wrong area, no error).

Five-minute pass over your diagram (Ch 4 · SOLID in interviews, Ch 1 · SOLID check)

Run S → O → L → I → D over the diagram you just drew. **Two findings is a good pass; five means rework; zero means you did not look.** S and O earn the most credit: O is what interviewers probe with a late requirement ("what if trucks are charged differently?" → "add a class").

Related ideas (Ch 4 · Cohesion, Coupling, SoC, SOLID Q&A)

Idea	Question it asks
SRP	How many actors can ask this class to change?
Cohesion	Do the parts of this class belong together?
Coupling	How much does A depend on B?
Separation of Concerns	Is each area of work in its own module?
SRP vs ISP	SRP splits **classes** (guards the worker); ISP splits **interfaces** (guards the caller)
DIP vs DI vs IoC	DIP = principle (depend on abstractions); DI = technique (pass dependencies in); IoC = the broader idea that a framework calls your code; IoC containers perform DI

Common mistakes: over-engineering (Ch 4)

Warning sign	Principle misapplied	Do instead
`XHelper`, `XUtils`, `XManager2` after a split	SRP	Split by actor; names must stand alone
Six small classes that always change in the same commit	SRP over-split	Merge them back
Interface + factory + registry for one implementation	OCP (*Speculative Generality*)	Write the `if`; refactor at the second or third case (rule of three)
A subclass that "works" by silently doing nothing	LSP	Better parent or composition
One interface per method	ISP over-split	Group by caller into role interfaces
An interface in front of stable `String`-like classes	DIP	Invert only volatile dependencies
A DI container for a 200-line program	DIP	Wire by hand in `main`

YAGNI is the counterweight: SOLID says *what* a good seam looks like; YAGNI says *when* to cut it.

In the interview

Say **observation → cost → fix → price of the fix**, and describe before you label: "`Reporting` implements nine methods but calls two — an interface segregation problem" (Ch 4 · SOLID in interviews).

22.5 Other design principles and Clean Code rules

Other principles (Chapter 5)

Principle	One-line idea	Smell it fixes	Key Java tool	Where
DRY	Each piece of knowledge has one authoritative home	Duplicated rules drifting apart	Value objects, constants, methods	Ch 5 · DRY
KISS	Choose the simplest design that works	Over-engineering	Plain code, `switch` expressions	Ch 5 · KISS
YAGNI	Don't build features before they are needed	Speculative Generality	Delete unused code	Ch 5 · YAGNI
Law of Demeter	Only talk to your immediate friends	Train wrecks `a.getB().getC().do()`	Intent-revealing delegate methods	Ch 5 · Law of Demeter
Tell, Don't Ask	Tell objects what to do; don't pull their data out and decide for them	Anemic model, Feature Envy	Command methods on the owner	Ch 5 · Tell, Don't Ask
Composition over inheritance	Prefer "has-a" to "is-a" for reuse	Fragile base class	Forwarding / decorators	Ch 5 · Composition over inheritance
Program to an interface	Depend on supertypes, choose the concrete class once at the edge	Hard-wired concrete dependencies	Interfaces + DI	Ch 5 · Program to an interface
Encapsulate what varies	Isolate the part that changes behind an interface	Growing type switches	Strategy, lambdas	Ch 5 · Encapsulate what varies
Loose coupling	Interacting objects know little about each other	God orchestrators	Observer, events, DI	Ch 5 · Loose coupling
Hollywood Principle / IoC	"Don't call us, we'll call you" — the framework owns the flow	Duplicated skeletons	Template Method, callbacks, DI	Ch 5 · Hollywood Principle
Separation of Concerns	Each part handles one concern	Spaghetti code	Layers, MVC	Ch 5 · Separation of Concerns, Ch 4
Least Astonishment	Do what the name promises and nothing more	Hidden side effects	Honest names, `Optional`	Ch 5 · Least Astonishment
Fail fast	Detect problems early and loudly	Distant `NullPointerException`s	Guard clauses, `requireNonNull`	Ch 5 · Fail fast
Immutability	Objects never change after creation	Aliasing, races	Records, `List.copyOf`	Ch 5 · Immutability
Design by Contract	Preconditions (caller), postconditions (method), invariants (always)	Magic error values	Checks at method entry/exit	Ch 5 · Design by Contract
GRASP	Nine guidelines for "which class should do this?"	God classes	Expert, Creator, Controller…	Ch 5 · GRASP
Domain-Driven Design	Model the business: bounded contexts, entities vs value objects, small aggregates, domain events, repositories per root	Anemic model, big ball of mud, primitive obsession	Records as value objects, aggregate roots, events + outbox	Ch 5 · DDD

GRASP in one line each (Ch 5 · GRASP):

Guideline	Rule
Information Expert	The class with the data does the work
Creator	The container (or the one with the initial data) creates
Controller	A thin non-UI object receives each system event and delegates
Low Coupling	Choose the design with fewer, more stable dependencies
High Cohesion	Each class does one related set of things
Polymorphism	Let each type implement the varying behaviour; don't switch on types
Pure Fabrication	Invent a helper class (e.g. a repository) when no domain class is a clean home
Indirection	Put a middleman between things that shouldn't know each other
Protected Variations	Wrap what's unstable behind a stable interface

Tensions to mention (Ch 5 summary): DRY vs loose coupling (sharing code couples its users); YAGNI vs encapsulate-what-varies (abstract only real or clearly predicted variation); Law of Demeter vs cohesion (too many delegates → Middle Man); Information Expert vs SoC (domain objects compute, fabrications persist).

Clean Code rules checklist (Chapter 6)

Area	Rule of thumb	Where
Names	Reveal intent; pronounceable; searchable; one word per concept; code reads like the requirement	Ch 6 · Meaningful names
Functions	Small; do one thing; one level of abstraction; top-down (stepdown); ≤ 2–3 arguments; **no flag arguments**; no hidden side effects; commands *or* queries (CQS); exceptions, not error codes	Ch 6 · Functions
Comments	Say it in code; comment only the *why*, warnings, legal text, public contracts; delete commented-out code	Ch 6 · Comments
Formatting	Newspaper order (headline first); related things close; one automated team style	Ch 6 · Formatting
Objects vs data structures	Objects hide data and expose behaviour; data structures expose data; no hybrids; no train wrecks	Ch 6 · Objects and data structures
Error handling	Unchecked exceptions with context; wrap vendor exceptions; empty collections / `Optional` / special case instead of `null`; never pass `null`; always clean up	Ch 6 · Error handling
Boundaries	Keep third-party code behind your own interfaces; learn it with tests	Ch 6 · Boundaries
Unit tests	Three laws of TDD; one concept per test; Given–When–Then; **F.I.R.S.T.** (Fast, Independent, Repeatable, Self-validating, Timely)	Ch 6 · Unit tests
Classes	Small by responsibility; cohesive; open for extension; depend on abstractions	Ch 6 · Classes
Systems	Build objects in one place (composition root) and inject them; cross-cutting concerns in proxies/aspects	Ch 6 · Systems
Simple design	Beck: passes the tests → reveals intention → no duplication → fewest elements (*Clean Code* swaps the middle two)	Ch 6 · Simple design
Concurrency	Separate concurrency code; share as little as possible; use the library; keep locks small; plan shutdown	Ch 6 · Concurrency
Smells	The C/E/F/G/J/N/T heuristics catalogue as a review vocabulary	Ch 6 · Smells and heuristics
Code review	Correctness → design → readability → tests → security → performance → operability; automate style; small PRs	Ch 6 · Code review checklist
Error strategy	Classify, translate at boundaries, retry only transient + idempotent with backoff, one global handler that logs once	Ch 6 · Error-handling strategy
Testing strategy	Many unit, some integration, few E2E; right test double; verify state over interactions; don't mock what you don't own	Ch 6 · Testing strategies

Boy Scout Rule: leave the code a little cleaner than you found it (Ch 6 intro).

22.6 Code smells → refactorings (all 24)

Source: Chapter 7 (code smells, refactoring catalogue, smell lookup table). ***Bold*** *= the usual first move.*

#	Smell	Signal (one line)	Group	Refactorings that cure it
1	Mysterious Name	Name doesn't say what it does/holds (`data`, `proc()`, `Manager`)	2nd-ed extra	**Change Function Declaration**, Rename Variable, Rename Field
2	Duplicated Code	Same structure in more than one place	Dispensables	**Extract Function**, Slide Statements, Pull Up Method
3	Long Function	You want a comment to explain a block	Bloaters	**Extract Function**, Replace Temp with Query, Introduce Parameter Object, Preserve Whole Object, Replace Function with Command, Decompose Conditional, Replace Conditional with Polymorphism, Split Loop
4	Long Parameter List	Many parameters, same types, boolean flags	Bloaters	Replace Parameter with Query, **Preserve Whole Object**, **Introduce Parameter Object**, Remove Flag Argument, Combine Functions into Class
5	Global Data	Readable and writable from anywhere (incl. mutable singletons)	2nd-ed extra	**Encapsulate Variable** (then shrink its scope)
6	Mutable Data	Shared data changed after creation	2nd-ed extra	**Encapsulate Variable**, Split Variable, Slide Statements, Extract Function, Separate Query from Modifier, Remove Setting Method, Replace Derived Variable with Query, Combine Functions into Class/Transform, Change Reference to Value
7	Divergent Change	One class changed for many different reasons (SRP)	Change Preventers	**Split Phase**, Move Function, Extract Function, **Extract Class**
8	Shotgun Surgery	One change forces many small edits in many classes	Change Preventers	**Move Function**, **Move Field**, Combine Functions into Class/Transform, Split Phase, Inline Function, Inline Class
9	Feature Envy	Method uses another class's data more than its own	Couplers	**Move Function**, Extract Function
10	Data Clumps	Same group of values always travel together (`startDate, endDate`)	Bloaters	**Extract Class**, **Introduce Parameter Object**, Preserve Whole Object
11	Primitive Obsession	Money as `double`, status as `String`	Bloaters	**Replace Primitive with Object**, Replace Type Code with Subclasses, Replace Conditional with Polymorphism, Extract Class, Introduce Parameter Object
12	Repeated Switches	The *same* switch on a type code in several places	OO Abusers	**Replace Conditional with Polymorphism**
13	Loops	Classic loop that filters/maps/aggregates	2nd-ed extra	**Replace Loop with Pipeline**
14	Lazy Element	A class/function that doesn't pull its weight	Dispensables	**Inline Function**, **Inline Class**, Collapse Hierarchy
15	Speculative Generality	Hooks "we might need someday"; abstract class with one subclass	Dispensables	Collapse Hierarchy, Inline Function, Inline Class, Change Function Declaration, Remove Dead Code
16	Temporary Field	Field set only in some situations, `null` otherwise	OO Abusers	**Extract Class**, Move Function, Introduce Special Case

#	Smell	Signal (one line)	Group	Refactorings that cure it
17	Message Chains	`order.getCustomer().getAddress().getCity()`	Couplers	**Hide Delegate**, Extract Function, Move Function
18	Middle Man	Most methods just forward to another class	Couplers	**Remove Middle Man**, Inline Function, Replace Superclass/Subclass with Delegate
19	Insider Trading	Two modules poke at each other's internals	Couplers	**Move Function**, **Move Field**, Hide Delegate, Replace Subclass/Superclass with Delegate
20	Large Class	Too many fields/methods — a God Class	Bloaters	**Extract Class**, Extract Superclass, Replace Type Code with Subclasses
21	Alternative Classes with Different Interfaces	Same job, different method names (`sendMail` vs `dispatch`)	OO Abusers	**Change Function Declaration**, Move Function, Extract Superclass
22	Data Class	Only fields and getters/setters; others manipulate it	Dispensables	**Encapsulate Record**, Remove Setting Method, Move Function, Extract Function, Split Phase
23	Refused Bequest	Subclass doesn't want inherited methods, or throws on them (LSP)	OO Abusers	Push Down Method/Field, **Replace Subclass with Delegate**, **Replace Superclass with Delegate**
24	Comments	Comment used as deodorant for unclear code	Dispensables	**Extract Function**, Change Function Declaration, Introduce Assertion

Smell family → principle violated (Ch 7 · Smell lookup table)

Smell family	Principle
Divergent Change, Large Class	SRP
Repeated Switches, Shotgun Surgery	OCP
Refused Bequest	LSP
Long Parameter List with flags, fat interfaces	ISP / clean function design
Global Data, Mutable Data	Encapsulation
Hard-wired `new` of concrete dependencies	DIP
Duplicated Code	DRY
Speculative Generality	YAGNI
Message Chains, Insider Trading	Law of Demeter
Feature Envy, Data Class	Tell, Don't Ask

Refactoring toward patterns (Ch 7)

Smell → pattern, in short (the full symptom table is in 22.7): algorithm-choosing switch → **Strategy**; methods checking and changing a status field → **State**; `if (x == null)` everywhere → **Null Object / Special Case**; sibling subclasses with the same skeleton → **Template Method**; many confusing constructors → **Factory Method** / creation methods; object built by many ordered calls → **Builder**; undo/queueing, or a huge function with shared locals → **Command**; logging/caching mixed into a core class → **Decorator**; polling or hard-coded notification → **Observer**; two classes, same job, different interfaces → **Adapter**; subclass explosion from feature combinations → **Strategy / Decorator / Bridge**; a pattern that no longer earns its keep → **remove it** (Inline Class, Collapse Hierarchy).

Refactoring = changing structure without changing behaviour, in small steps, with tests as the safety net. Wear one of **two hats** at a time: adding function or refactoring (Ch 7 · Refactoring fundamentals).

22.7 Design patterns: all 23 GoF + extras

Source: Chapters 8–11 (catalogue in Ch 8 · The 23 GoF patterns; comparison tables at the end of Chapters 8, 9, 10 and 11).

Count: 5 creational + 7 structural + 11 behavioral = **23**. **Class-scoped** (inheritance) patterns: Factory Method, class Adapter, Template Method, Interpreter; almost everything else is object-scoped (composition) (Ch 8 · The 23 GoF patterns).

Creational (Chapter 8)

Pattern	Intent (one line)	Use when	Java / JDK example	Commonly confused with	Where
Singleton	Exactly one instance with a global access point	There really must be exactly one (config, pool, registry)	`Runtime.getRuntime()`; Spring singleton-scoped beans	A global variable; a DI singleton scope (preferred for testing)	Ch 8
Factory Method	Subclasses decide which concrete class to create via an overridable method	Shared procedure, varying product; others will add types	`Collection.iterator()`	Simple Factory (one method with a switch)	Ch 8
Abstract Factory	Create *families* of related objects without naming concrete classes	Products must match; swap a whole set at once	`DocumentBuilderFactory`; JDBC `Connection`	Factory Method (one product by inheritance vs a family by composition)	Ch 8
Builder	Build a complex object step by step; one validating `build()`	Many optional fields, cross-field validation, immutable result	`StringBuilder`, `HttpClient.newBuilder()`	A plain constructor (fine for a handful of required fields); telescoping constructors	Ch 8
Prototype	Create new objects by copying a configured instance	Setup is expensive, or only the interface is known	Copy constructors, `ArrayList.clone()`	Memento (a snapshot to restore, not a copy to use)	Ch 8

Singleton notes (Ch 8 · TL;DR): eager, holder-class or **`enum`** (most robust: blocks reflection and serialization duplicates); double-checked locking needs a **`volatile`** field. It is global state that hurts testing — prefer injecting one shared instance. Many case studies reject it (Ch 14 self-check).

Structural (Chapter 9)

Pattern	Intent (one line)	Use when	Java / JDK example	Commonly confused with	Where
Adapter	Convert one interface into another that clients expect	"The vendor API doesn't match ours" (after the fact)	`Arrays.asList`, `InputStreamReader`	Bridge (same shape, but Bridge is designed up front); Facade	Ch 9 · Adapter Pattern
Bridge	Split abstraction from implementation so both vary independently	Class names with two nouns; two dimensions multiplying classes	JDBC `Connection` + drivers	Adapter; Strategy (same shape)	Ch 9 · Bridge Pattern
Composite	Treat single objects and trees uniformly	Part-whole trees (files, menus, UI)	`java.awt.Container` is a `Component` holding `Component`s	Decorator (also wraps a component, but adds behaviour to one)	Ch 9 · Composite Pattern
Decorator	Add responsibilities dynamically by wrapping; same interface, stackable	"Sometimes logging, sometimes caching, sometimes both"	`BufferedInputStream`, `Collections.synchronizedList`	Proxy (same shape; Proxy *controls access*); Chain of Responsibility (in CoR someone can stop the flow)	Ch 9 · Decorator Pattern
Facade	One simple entry point to a complex subsystem	Every caller repeats the same multi-step sequence	SLF4J `LoggerFactory`, service layers	Adapter (converts one object's interface); Mediator	Ch 9 · Facade Pattern
Flyweight	Share intrinsic (immutable) state across huge numbers of objects	Millions of near-identical objects	`Integer.valueOf`, `String.intern()`	Object Pool (reuses objects one borrower at a time; flyweights are shared at once)	Ch 9 · Flyweight Pattern
Proxy	A stand-in that controls access (lazy, remote, protective, caching)	It's expensive, remote or restricted	`java.lang.reflect.Proxy`, Spring `@Transactional`, Hibernate lazy load	Decorator	Ch 9 · Proxy Pattern

Memory hooks (Ch 9 · Structural comparison): Adapter *make it fit* · Decorator *add to it* · Proxy *guard it* · Facade *simplify it* · Bridge *split it* · Composite *nest it* · Flyweight *share it*.

Behavioral (Chapters 10–11)

Pattern	Intent (one line)	Use when	Java / JDK example	Commonly confused with	Where
Chain of Responsibility	Pass a request along handlers until one handles (or stops) it	Several checks, any may reject; handler not known in advance	Servlet `Filter` chain; `java.util.logging` handlers	Decorator (every layer always runs)	Ch 10
Command	Turn a request into an object: queue, log, retry, undo	Undo/redo, job queues, scheduling	`Runnable` / `Callable`; Swing `UndoManager`	Strategy (Strategy = *how*; Command = *what*)	Ch 10
Iterator	Traverse a collection without exposing how it is stored	Walk a collection/tree/paged API	`Iterator`, `Spliterator`, JDBC `ResultSet`	Visitor (what to do at each node)	Ch 10
Mediator	Centralise communication between peers in one coordinator	Many objects interact in tangled ways; order of reactions matters	Chat servers, UI controllers; `Executor` / `Timer` are a loose fit (Ch 10)	Observer (broadcast, no coordination); Facade	Ch 10
Memento	Capture and restore state without breaking encapsulation	Undo, checkpoints, rollback	A `record` snapshot; DB `ROLLBACK`, Git commits (real-world)	Command undo (store actions, not states); Prototype	Ch 10
Interpreter	One class per grammar rule; evaluate sentences of a small language	User-written rules/expressions	`java.util.regex.Pattern`, SpEL	Composite (same tree); Visitor (many operations over the AST)	Ch 10
Observer	Notify many dependents automatically when one object changes	An open-ended set of independent listeners	`PropertyChangeListener`, `java.util.concurrent.Flow`, Spring `@EventListener`	Mediator; pub-sub (Observer through a broker, Ch 20 · Pub-sub)	Ch 11
State	Behaviour changes with internal state, as if the class changed	Many states/transitions; methods repeat a `switch` on status	TCP connection (GoF's example), order lifecycles. `Thread.State` is only an enum of states, not the pattern	Strategy (same diagram; states know each other and advance themselves)	Ch 11
Strategy	Make a family of algorithms interchangeable behind one interface	Several ways to do one job, chosen at run time	`Comparator`; `RejectedExecutionHandler` policies	State; Template Method; Decorator (Strategy *replaces* a rule, Decorator *wraps* one)	Ch 11
Template Method	Fixed algorithm skeleton in a base class; subclasses fill in steps	Long fixed flow, a few varying steps	`AbstractList`, `InputStream.read(byte[],…)`, `HttpServlet.service()`	Strategy (Template = inheritance, one step varies;	Ch 11

Pattern	Intent (one line)	Use when	Java / JDK example	Commonly confused with	Where
				Strategy = composition, whole algorithm varies)	
Visitor	Add operations to a stable set of classes without changing them (double dispatch)	Many new operations over a fixed type set (ASTs)	`javax.lang.model.element.ElementVisitor`; `FileVisitor` is named after it but is really a tree-walk callback	Iterator; in Java 21, **sealed types + pattern-matching** `switch` usually replace it	Ch 11

Beyond GoF: extras you will meet

Pattern	Intent (one line)	Use when	Java / real example	Commonly confused with	Where
Simple Factory (idiom)	One `create(type)` method maps a type value to a class	Small, stable set of types chosen by config	`Calendar.getInstance()`, `EnumSet.noneOf`	Factory Method	Ch 8
Object Pool	Borrow and return a bounded set of expensive objects	DB connections, limited resources	HikariCP, `ThreadPoolExecutor`	Flyweight	Ch 8
Null Object	A do-nothing implementation instead of `null`	`null` checks everywhere for an optional collaborator	`Collections.emptyList()`, `OutputStream.nullOutputStream()`	`Optional` (for lookups)	Ch 11 · Beyond GoF
Dependency Injection	Receive collaborators (as interfaces) through the constructor	Classes create their own collaborators; hard to test	Spring constructor injection; a hand-wired `main`	Service Locator (anti-pattern: the class asks for its dependencies)	Ch 11 · Beyond GoF
Repository	Collection-like interface over storage	Business logic mixed with storage code	Spring Data `CrudRepository`	DAO (table-oriented)	Ch 11 · Beyond GoF
Specification	A named, combinable business rule (`isSatisfiedBy`, and/or/not)	Same rule in queries and validation; finder-method explosion	`Predicate`; Spring Data JPA `Specification`	—	Ch 11 · Beyond GoF
MVC	Separate model, view and controller (Observer + Strategy + Composite)	UI, input handling and rules tangled	Spring MVC; Swing `JTable` + `TableModel`	MVP, MVVM (relatives)	Ch 11 · Beyond GoF
Event Sourcing (lite)	Store the events; derive current state by replaying them	Full history, audit, time travel	Append-only logs	Plain audit log	Ch 11 · Beyond GoF
Architectural styles (layered, client-server, modular monolith vs microservices, event-driven, pipe-and-filter, hexagonal/clean)	Fix the shape of the *whole system*: parts, dependency direction, communication	Deciding where classes live and answering "would you split this?"	Spring layered apps; Kafka consumers; Unix pipes; ports and adapters	Design patterns (a few classes inside one part); MVC (only the UI edge)	Ch 11 · Architectural patterns

"Which pattern?" by problem symptom

Combined from Ch 8 · How to pick a pattern, Creational patterns compared, Ch 9 · Structural comparison, Ch 11 · Choosing a behavioral pattern and Ch 7 · Refactoring toward patterns.

If you see / need…	Use	Not to be confused with
One operation, several interchangeable rules chosen at run time (pricing, matching, routing)	**Strategy**	State, Template Method
The class to create depends on a type value; caller shouldn't know the concrete class	**Factory** (Simple Factory / Factory Method)	Abstract Factory
Products must always match as a family	**Abstract Factory**	Factory Method
Constructor has ~6+ arguments, many optional	**Builder**	—
Another one of *this*, cheaply or without knowing its class	**Prototype**	Memento
Exactly one instance must exist	**Singleton** (or DI singleton scope)	A convenient global handle
Expensive objects created and destroyed repeatedly	**Object Pool**	Flyweight
Existing class has the wrong interface	**Adapter**	Facade, Bridge
Two dimensions multiplying classes (`WindowsButton`, `MacButton`, …)	**Bridge**	Adapter
Tree of part-whole objects treated the same	**Composite**	Decorator
Add behaviour in combinations without editing the class	**Decorator**	Proxy
Every caller repeats the same five steps into a subsystem	**Facade**	Mediator
Millions of near-identical objects	**Flyweight**	Object Pool
Lazy load, access control, remote, caching (same interface)	**Proxy**	Decorator
A request must pass several checks; any may reject it	**Chain of Responsibility**	Decorator (all layers run)
Undo/redo, queue, schedule, retry or log *requested actions*	**Command**	Strategy
Evaluate user-written rules in a small language	**Interpreter** (+ Specification)	Visitor
Walk a collection/tree/paged API without exposing structure	**Iterator**	Visitor
Many objects depend on each other and the **order** of reactions matters	**Mediator**	Observer
Roll back to an earlier state without exposing internals	**Memento**	Command undo
One change must reach an open-ended set of listeners	**Observer** (or a broker)	Mediator
Three or more methods repeat one `switch` over a status; legal actions depend on it	**State** (or enum + guarded method for 3–4 states)	Strategy
Several classes share a long fixed flow, differ in a few steps	**Template Method**	Strategy
Many new operations over a stable set of types	**Visitor**, or sealed types + pattern matching	Iterator
`null` checks everywhere	**Null Object** / `Optional`	—
Classes create their own collaborators	**Dependency Injection**	Service Locator
Business logic mixed with storage	**Repository**	DAO
Same rule duplicated in queries and validation	**Specification**	—

Three questions for the famous look-alikes (Ch 11 · Choosing a behavioral pattern):

1. Do the alternatives reference each other? Yes → **State**. No → **Strategy** (or Bridge if chosen once at wiring).
2. Can one participant stop the flow? Yes → **Chain of Responsibility**. Everyone always runs → **Decorator**. Everyone is told, nobody stops anyone → **Observer**.

3. Does a central object decide what happens next? Yes → **Mediator**. Each listener decides for itself → **Observer**.

The six that come up in almost every round (Ch 1 · Patterns in this round, Ch 8 · How to pick a pattern)

Pattern	Signal	Do not use when
Strategy	One operation, several swappable rules	Only one rule exists and nothing hints at a second
Factory	Creation depends on a type value	Only one concrete class exists
Observer	Several listeners the source should not know	A single listener the source already references
State	Behaviour depends on status; many transitions	3–4 states handled by one guarded method
Singleton	Genuinely exactly one	You just want a global handle
Builder	Many optional fields; long constructor	A handful of required fields

Common mistakes (Ch 8 · Anti-patterns)

Anti-pattern	Fix
God Object (one class does everything)	Split by responsibility; push behaviour to the data owners
Golden Hammer (favourite pattern everywhere)	Start from the problem's signal
Pattern overuse ("patternitis") — factories for one class, single-implementation interfaces	YAGNI: add the pattern when the second variant arrives
Singletonitis (singletons as globals)	Dependency injection
Cargo-cult patterns (structure without intent)	Name the property the pattern gives you, or drop it

In the interview

Use **one or two** patterns in a 45-minute design, placed where their signal appears, and **name one you rejected** and why (Ch 8 · How to pick a pattern, self-check).

22.8 Concurrency cheat sheet

Source: Chapter 12 (concepts) and Chapter 13 (classic problems).

TL;DR (Ch 12)

Races come from **shared mutable state**. Make the critical section atomic, or remove sharing/mutability. Visibility needs **happens-before** edges. Prevent deadlock with **lock ordering** or **tryLock + back-off**. In LLD: name the contended resource, keep **check + claim atomic**, lock fine-grained and in order, and say what changes with many servers.

Primitive | use for | gotcha

Primitive	Use for	Gotcha	Where
`synchronized`	Making a multi-step update atomic (mutex + visibility)	Don't lock on a changing reference, a boxed value or a string literal; no slow I/O while holding it; readers must use the same lock	Ch 12 · Mutex and synchronized
`ReentrantLock`	Same as `synchronized` plus try/timed/fair locking and `Condition`s	Call `lock()` **just before** `try`, `unlock()` in `finally`; fairness is noticeably slower	Ch 12 · ReentrantLock and reentrancy
`tryLock(timeout)`	Give up if the lock is busy; deadlock prevention with back-off	Never `unlock()` when `tryLock` returned `false`; don't retry in a tight loop (livelock)	Ch 12 · Try-lock & timed locking
`ReentrantReadWriteLock`	Many readers, few writers	Cannot upgrade read → write (self-deadlock); don't modify under the read lock; writer starvation in non-fair mode	Ch 12 · Read-write locks
`StampedLock`	Read-mostly data with optimistic reads	**Not reentrant**; `validate` the stamp before using values read optimistically	Ch 12 · Read-write locks
`volatile`	Visibility of a flag or a published immutable snapshot	Gives visibility, **not** atomicity: `count++` is still a race	Ch 12 · Java Memory Model
`AtomicInteger` / `AtomicLong` / CAS	Lock-free counters and single-variable claims	Separate `get()` then `set()` is still a race — use `compareAndSet` / `updateAndGet`; no side effects in the lambda (may run several times); ABA → `AtomicStampedReference`	Ch 12 · CAS & atomics
`LongAdder`	Very hot counters (metrics)	Use instead of `AtomicLong` when contention is high	Ch 12 · CAS & atomics
`Semaphore(N)`	Limit concurrent access to N; signalling between threads	Release in `finally`; don't release when `acquire` never succeeded	Ch 12 · Semaphores
`wait/notifyAll`, `Condition`	Waiting for a condition on shared state	Always `while`, never `if` (spurious wakeups); hold the monitor; `notify()` can wake the wrong thread; state change without notify = lost wakeup	Ch 12 · Condition variables
`CountDownLatch`	Wait for N events once	One-shot; `countDown()` in `finally`, `await(timeout)`	Ch 12 · CountDownLatch and CyclicBarrier
`CyclicBarrier` / `Phaser`	N threads meet at each phase (reusable)	Fewer threads than parties → everyone waits forever	Ch 12 · CountDownLatch and CyclicBarrier
`ConcurrentHashMap`	Shared map	`containsKey` then `put` is a race — use `putIfAbsent` / `computeIfAbsent` / `merge`; no slow work or recursive updates inside `compute`; no `null` keys/values	Ch 12 · Concurrent collections
`CopyOnWriteArrayList`	Read-mostly lists (listener lists)	Every write copies the array	Ch 12 · Concurrent

Primitive	Use for	Gotcha	Where
			collections
`BlockingQueue` (bounded)	Handing work between threads (producer–consumer)	Unbounded queue → no back-pressure → memory blow-up; one poison pill per consumer	Ch 12 · Producer-consumer
`ExecutorService` / `ThreadPoolExecutor`	Running tasks on a bounded pool	Call `shutdown()`; `newFixedThreadPool` has an **unbounded** queue; exceptions in `submit`ted tasks are lost unless you call `get()`; tasks waiting on tasks in the same pool → thread-starvation deadlock	Ch 12 · Creating threads, Thread pool pattern
Virtual threads	Many blocking I/O tasks	Pinning (Java 21–23: blocking inside `synchronized`); heavy `ThreadLocal`s waste memory	Ch 12 · Creating threads, Immutability & confinement
`ForkJoinPool` / parallel streams	Splitting CPU work	Use a threshold: tiny tasks cost more to schedule than to run; subtasks must touch disjoint data	Ch 12 · Choosing the tool, Ch 13 · Parallel merge sort
`CompletableFuture`	Combining async results (`thenApply`, `thenCombine`, `allOf`)	Handle errors with `exceptionally` rather than calling `get()` at every step	Ch 12 · Creating threads
`ThreadLocal`	Per-thread state (thread confinement)	Clear it (`remove()`) in pooled threads, or values leak into the next task	Ch 12 · Immutability & confinement
Immutability / confinement	Avoiding synchronization entirely	`final List` is not immutable if the list is mutable and exposed	Ch 12 · Immutability & confinement

Which `java.util.concurrent` class for which job (Ch 12 · Choosing the tool)

Counter → `AtomicInteger`/`AtomicLong` (`LongAdder` if very hot) · stop a loop from another thread → `volatile boolean` or `interrupt()` · atomic multi-step update → `synchronized`/`ReentrantLock` · give up if busy → `tryLock(timeout)` · many readers, few writers → `ReentrantReadWriteLock`, `StampedLock` or an immutable snapshot in a `volatile` field · limit access to N → `Semaphore(N)` · wait for a condition → `Lock` + `Condition` (or `wait/notifyAll`) in a `while` loop · wait for N events once → `CountDownLatch` · N threads meet each phase → `CyclicBarrier`/`Phaser` · hand work between threads → bounded `BlockingQueue` · shared map → `ConcurrentHashMap` with `putIfAbsent`/`compute`/`merge` · read-mostly list → `CopyOnWriteArrayList` · run tasks → bounded `ExecutorService`, virtual threads for blocking I/O · split CPU work → `ForkJoinPool`/parallel streams · combine async results → `CompletableFuture` · avoid synchronization → immutability, thread confinement.

Deadlock checklist (Ch 12 · Deadlock, Livelock, Starvation)

A deadlock needs **all four Coffman conditions**; break any one and it cannot happen.

#	Condition	How to break it
1	Mutual exclusion	Lock-free/immutable data, shareable resources (read locks)
2	Hold and wait	Acquire everything at once, or `tryLock` and release everything on failure
3	No preemption	Timeouts (`tryLock(timeout)`), leases that expire
4	Circular wait	**Global lock ordering** (always lock the lower id first)

- ⬜ Do I ever hold two locks? If yes, is there **one global order** (account id, seat id), with a tie-breaker for equal keys?

- ☐ Do I call a **listener, callback or another object's synchronized method while holding my lock**? Move it outside the lock.
- ☐ Do tasks in a bounded pool **wait for other tasks in the same pool**? (thread-starvation deadlock)
- ☐ Are retries randomised (**jitter**) so two threads don't back off in lockstep? (livelock, Ch 12)
- ☐ Could one thread wait forever while others progress? Consider fairness (Ch 12 · Starvation).
- ☐ Detection: a thread dump (`jstack` / `jcmd`) prints "Found one Java-level deadlock"; `ThreadMXBean.findDeadlockedThreads()`.

Deadlock vs livelock vs starvation: nobody progresses and all wait / everybody runs but nobody progresses / some progress while one never does (Ch 12 · self-check).

The LLD concurrency recipe (Ch 12 · Concurrency in LLD interviews)

1. **Name the shared mutable state** (free spots, seat → owner, stock).
2. **State the invariant** ("a seat has at most one booking", "stock ≥ 0").
3. **Find the critical operation** — usually a check-then-act (*find free → assign*).
4. **Pick the smallest tool that makes it atomic:** one item → `queue.poll()`, `putIfAbsent`, `compareAndSet`; several items all-or-nothing → per-item locks **in sorted order**; across servers → DB constraints/transactions, optimistic versioning, or a distributed lock.
5. **Choose granularity:** per floor / show / SKU, not one global lock.
6. **Handle time:** holds with expiry, timeouts, idempotent retries.

Classic problems (Ch 13 · summary)

Problem	Key primitive	Invariant to state
FooBar (Ch 13 · Print FooBar Alternately)	2 semaphores	Exactly one permit exists across both
Zero Even Odd (Ch 13 · Print Zero Even Odd)	3 semaphores	Zero dispatches; one permit circulates
FizzBuzz (Ch 13 · Fizz Buzz Multithreaded)	monitor + `notifyAll`	Predicates partition 1..n
H2O (Ch 13 · Building H2O)	2 semaphores + barrier	≤ 2 H and ≤ 1 O per round
Dining Philosophers (Ch 13)	ordered locks / n−1 semaphore	No circular wait
TTL cache (Ch 13 · Thread-safe cache with TTL)	`ConcurrentHashMap` + `compute` + conditional remove	Never return or keep expired data
Striped map (Ch 13 · Striped concurrent map)	lock array	Each bucket belongs to exactly one stripe
Blocking queue (Ch 13 · Thread-safe bounded Blocking Queue)	lock + 2 conditions	0 ≤ count ≤ capacity; wait in loops
Bloom filter (Ch 13 · Concurrent Bloom Filter)	atomic OR	Bits only go 0 → 1; no false negatives
Merge sort (Ch 13 · Parallel merge sort)	Fork/Join	Subtasks touch disjoint ranges
Token-bucket rate limiter (Ch 13)	lock or CAS on immutable state	Refill + take is one atomic step

Common mistakes (Chapter 12)

- A global `synchronized` on `ParkingLot.park()` — correct, but serialises every gate.
- Checking availability with one call and booking with another (check-then-act).
- Locking seats in the order the user clicked them (deadlock between two multi-seat bookings).
- Forgetting that in-memory locks don't work across multiple app servers.
- Swallowing `InterruptedException` — rethrow, or restore the flag with `Thread.currentThread().interrupt()` (Ch 12 · Thread states).

- Calling `run()` instead of `start()` (Ch 12 · Thread states).

22.9 Case-study map

Source: Chapters 14–21. "Concurrency concern" is the race each chapter guards against, and "atomic step" is how.

Everyday objects (Chapter 14)

Problem	Key entities	Patterns used	Concurrency concern	Where
Parking lot	`ParkingLot`, `ParkingFloor`, `ParkingSpot`, `Vehicle`, `Ticket`, `DisplayBoard`, `SpotType` / `VehicleType` enums	Strategy (pricing, spot-finding), Factory (electric spots), Observer (boards); Singleton rejected	Two gates take one spot → atomic `pollFirst` + CAS; per-ticket lock for payment; `putIfAbsent` for plates	Ch 14
Vending machine	`VendingMachine`, `Slot`, `Inventory`, `CashBox`, `Denomination`, `VendingState` + 5 states	**State** (the whole design)	`synchronized` machine; change-making by bounded coin-change DP, not greedy	Ch 14
Coffee machine	`CoffeeMachine`, `Recipe`, `RecipeBook`, `IngredientStore`, `OrderResult` (sealed)	Observer (low stock), recipes as data	Two orders take the last milk → one `ReentrantLock` around check + deduct; notify outside the lock	Ch 14 · Coffee Vending Machine
ATM	`Atm`, `AtmState` + 4 states, `Transaction` (sealed), `NoteHandler`, `CashDispenser`, `BankService`, `Account`	State (session), Command (transactions), Chain of Responsibility (note cassettes)	Overdraw from two ATMs → per-account lock; transfers lock in id order	Ch 14
Elevator	`Elevator`, `StopList` (two sorted sets), `Door`, `Dispatcher`, `DispatchStrategy`, `Building`	Strategy (dispatch) + modes; LOOK scheduling	Per-car lock, synchronized dispatcher; tick-driven `step()`	Ch 14 · Elevator System
Traffic signal	`Intersection`, `TrafficLight`, `Phase`, `SignalState` + 5 states, `TimingPolicy`, `SafetyMonitor`	State (timed), Strategy (timing), Observer (lights)	Synchronized tick; safe with a timer thread plus emergency threads	Ch 14 · Traffic Signal Control System
Shipping locker (Amazon Locker)	`LockerStation`, `Compartment`, `Booking`, `CompartmentSelector`, `PickupResult` (sealed), `LockerListener`, `LockerNetwork`	Strategy (door choice), Observer (SMS), enum state machines, sealed results	Two carriers, one door → station lock around choose-and-mark; lazy expiry of holds and pickup windows; hashed single-use pickup and return drop-off codes, keypad lockout	Ch 14 · Shipping Locker System

Booking & inventory (Chapters 15–16)

Problem	Key entities	Patterns used	Concurrency concern	Where
Library	`Book` (title) vs `BookItem` (copy), `Member`, `BookLending`, `LendingService`, `Catalog`	Strategy (fines), notifier interface; FIFO reservation queue per title	Last copy lent twice → lock member then copy (fixed order)	Ch 15 · Library
Movie tickets (BookMyShow)	`Show`, `ShowSeat`, `Booking`, `SeatLockProvider`, `BookingService`, `CancellationPolicy`	Strategy (refund rule), gateway behind an interface	Book `ShowSeat`s; all-or-nothing hold with TTL → pay → atomic commit; refund if hold lost	Ch 15 · BookMyShow
Concert tickets	`SeatedInventory`, `GaPool`, `SeatState` (sealed), `Hold`, `VirtualWaitingRoom`, `PurchaseLimiter`	Sealed seat states, waitlist	Extreme contention → CAS per seat, CAS counter for GA, FIFO waiting room, per-user limits, TTL holds with a reaper	Ch 15 · Concert tickets
Hotel	`Hotel`, `Room`, `RoomBooking`, `Invoice`, `RoomCharge`, `Payment` subclasses, `FrontDesk`	Template Method (`RoomCharge`)	Room × night reserved atomically in `synchronized Room.tryReserve`	Ch 15 · Hotel
Car rental	`Vehicle`, `DateRange`, `Reservation`, `Bill` / `BillItem`, `ReservationAddOn`, `PricingStrategy`	Template Method (add-ons), Strategy + Decorator (pricing), state machine	Availability is about **dates** → `Vehicle.tryBook(range)` synchronized	Ch 16 · Car Rental
Restaurant	`Menu` / `MenuSection` / `MenuItem`, `Table`, `Reservation`, `Order`, `Meal` / `MealItem`, `Kitchen`, `Check`	Observer (order ready), Strategy (chef assignment, payment), state machine	Two receptionists, one table → `Table.tryHold(time)` synchronized; listeners after the lock	Ch 16 · Restaurant
Airline	`Flight` vs `FlightInstance`, `Seat` vs `FlightSeat`, `FlightReservation`, `Itinerary`, `FareStrategy`	Strategy + Decorator (fares), Facade	Selling one seat twice → `FlightInstance.holdSeats` all-or-nothing under the instance lock	Ch 16 · Airline
Course registration	`Course` vs `Section`, `Student`, `RegistrationRule`s, `Registrar`	Chain of rules / Specification	Overselling a section → section lock inside a student lock (fixed order)	Ch 16 · Course Registration
Meeting scheduler	`Interval` (half-open), `Room` (owns a `TreeMap` calendar), `Meeting`, `Invitation`, `RoomSelectionStrategy`, `MeetingObserver`, `Recurrence`	Strategy (room ranking), Observer (invites, RSVPs)	Two organizers, overlapping times → check (`lowerEntry`) and insert under the room lock; rank then try each room; lock order meeting → room	Ch 16 · Meeting Scheduler

Games (Chapter 17)

Problem	Key entities	Patterns used	Concurrency concern	Where
Blackjack + deck	`Card`, `Deck<T>`, `Shoe`, `Hand`, `BlackjackScoring`, `PlayerStrategy`, `DealerPolicy`	Strategy (player, dealer); injected randomness	One table = one thread; shared balances need an atomic wallet service	Ch 17 · Blackjack
Chess	`Piece` (sealed) + subclasses, `SlidingPiece`, `Board`, `Move`, `Rules`, `Game`	Polymorphic pieces, Factory Method, Command + Memento (undo)	`synchronized` game so a move and a resignation don't interleave	Ch 17 · Chess
Tic-tac-toe	`Board` (row/col/diagonal counters), `Game`, `Player`, `MoveStrategy`	Strategy (players)	Many games in a `ConcurrentHashMap`; per-game lock	Ch 17 · Tic-Tac-Toe
Snake & ladder	`Board`, `Jump`, `Dice`, `Rules`, `OverflowRule`, `WinStrategy`, `Game`	Strategy (dice, overflow, win)	Each game its own `Random` and its own thread of turns	Ch 17 · Snake and Ladder
2048	`LineMerger`, `Board`, `TileSpawner`, `Game`, `Direction`	Strategy (spawner); one pure merge function	Single-threaded; `synchronized move` if UI and input threads both call it	Ch 17 · 2048
Jigsaw puzzle	`Side` (FLAT/TAB/BLANK + shape code), `Piece` (immutable), `Board` + `Placement` (piece, rotation), `PuzzleValidator`, `SolverStrategy`	Strategy (brute force vs edge-first with a side index); one placement check for players and solvers	Co-op players → `synchronized` game; solvers carry an attempt budget	Ch 17 · Jigsaw

Platforms (Chapters 18–19)

Problem	Key entities	Patterns used	Concurrency concern	Where
Stack Overflow	`Post` (sealed) → `Question` / `Answer` / `Comment`, `Member`, `Tag`, `Bounty`, `QuestionIndex`	Strategy (reputation), Observer (badges), Facade; inverted index	Two votes at once → `synchronized` post; `AtomicInteger` reputation	Ch 18
Cricinfo	`Match` → `Innings` → `Over` → `Ball`, `Team`, `Playing11`, `StatQuery`	Template Method (stat queries), Observer (live pages)	Two scorers → `synchronized` match; observers in `CopyOnWriteArrayList`	Ch 18
LinkedIn	`Member`, `Profile`, `SocialContent`, `ConnectionInvitation`, `ConnectionService`, `SearchIndex`, `FeedService`	State (invitation), Strategy (feed ranking, notifications), Facade	Symmetric connections → all graph changes synchronized on the service	Ch 18
Facebook	`Member`, `Audience` (sealed), `PrivacyList`, `FriendService`, `VisibilityService`, `NewsFeedService`	Strategy (feed ranker), Observer (notifications)	Accept can't leave a one-way friendship → synchronized service, concurrent adjacency sets	Ch 18
Amazon	`Product`, `ShoppingCart`, `Order`, `OrderLog`, `Payment`, `Shipment`, `Customer` → `Guest` / `Member`	Strategy (payment), Observer (notifications), order state machine, Facade	Last unit, two buyers → `synchronized Product.reserve`; all-or-nothing multi-line reservation	Ch 19 · Amazon
Stock brokerage + exchange	`Order` → Market/Limit/StopLoss/StopLimit, `OrderPart`, `StockLot`, `OrderBook`, `MatchingEngine`	Observer (fills), Template Method (statements), trade-price policy	Over-spending → reserved funds; one lock per symbol; price-then-time matching	Ch 19 · Stock Brokerage
Auction	`AuctionListing`, `Bid`, `BidResult`, `AuctionService`, `AuctionListener`	Observer, Specification (search), status state machine	Concurrent bids → all bid rules in one per-listing critical section; notify after unlocking	Ch 19 · Auction
Music streaming (Spotify)	`Song` / `Album` / `Artist`, `MusicLibrary`, `Playlist`, `MusicPlayer`, `PlaybackOrder`, `Recommender`	Strategy (play order, recommendations), Observer (plays), plan rules in an enum	Two devices press "next" → `synchronized` player methods	Ch 19 · Music Streaming
Food delivery (Swiggy) + FoodKart	`Restaurant`, `Cart`, `Order`, `DeliveryAgent`, `DeliveryAssignmentStrategy`, `RestaurantSelectionStrategy`	Strategy (assignment, selection, payment), Observer, order state machine	Kitchen capacity reserved atomically; riders claimed with CAS	Ch 19 · Food Delivery
Grocery store (POS)	`Register`, `Sale`, `LineItem`, `PricingEngine`, `ItemPromotion` / `BasketPromotion`, `TaxPolicy`, `Tender` (sealed), `Receipt`, `Inventory`	Strategy (offers, tax, payment gateway), Facade (register), Observer (stock), sale state machine	Registers share stock → atomic `merge`, may go negative, edge-triggered alerts; returns claimed with `compute`	Ch 19 · Grocery Store

Pattern-driven components (Chapter 20)

Problem	Key entities	Patterns used	Concurrency concern	Where
Splitwise	`Expense`, `SplitStrategy` (Equal/Exact/Percent), `Share`, `Ledger`, `DebtSimplifier`	Strategy (splits)	One signed number per pair, one mutator; `BigDecimal`, no lost cents	Ch 20
Notification service	`NotificationChannel` (abstract), channels, `SendRule`, `ChannelFactory`, `DeliveryListener`	Template Method (channels), Strategy (send rules), Observer, Simple Factory, Adapter	Rate-limit check-then-act gap; idempotency	Ch 20
In-memory cache / LRU	`Cache<K,V>`, `Node`, `DoublyLinkedList`, `EvictionPolicy`, `Cache.Builder`, `ShardedCache`	Strategy (eviction), Observer (removals), Builder	One lock (a `get` is a write in LRU); listeners outside the lock; shard to scale	Ch 20
Ride sharing (Uber)	`Rider`, `Driver`, `Trip`, `MatchingStrategy`, `PricingStrategy`, `SurgePricing`	Strategy ×3, Decorator (surge), trip state machine	Double-booking a driver → atomic claim (`AtomicBoolean`)	Ch 20 · Ride sharing
Logging framework	`Logger`, `LogManager`, `LogRecord`, `Appender`, `AsyncAppender`, `Formatter`	Chain of Responsibility (hierarchy), Singleton (`LogManager`), Strategy (formatters), Template Method (appender), Decorator (async)	Thread-safe sinks; never block on a full queue; never crash the app	Ch 20
Pub-sub	`Broker`, `Topic`, `Subscription`, `Subscriber`, `Message`	Observer through a broker	One dispatcher per subscription; ack after processing (at-least-once) → idempotent subscribers	Ch 20 · Pub-sub
Rate limiter	`Limiter` (TokenBucket, FixedWindow, SlidingWindowLog/Counter), `CompositeLimiter`, `LimiterFactory`	Strategy, Simple Factory, Composite, Facade	Atomic refill-check-take; one limiter per key via `computeIfAbsent`; Redis + Lua across servers	Ch 20

Machine-coding problems (Chapter 21)

Problem	Key entities	Patterns used	Concurrency concern	Where
Task management	`Task`, `TaskStatus`, `TaskService`, `TaskRepository`, `TaskFilter`	Specification (filters), Strategy (sorts), Observer, enum state machine	Per-task `synchronized`	Ch 21 · Task management
Digital wallet	`Wallet`, `Transaction`, `Money`, `Offer`, `WalletService`	Strategy (offers), idempotency key	Lock both wallets in a global (id) order; debit before credit; total conserved	Ch 21 · Digital wallet
Event calendar (Flipkart)	`TimeSlot` (half-open), `Event`, `EventStore`, `RepresentativeSelector`, `CalendarService`	Strategy (representative), plan-then-save	Atomic check-and-book	Ch 21
Coding Blox (Flipkart)	`Contest`, `Level`, `User`, `Question`, `SolveSimulator`, `CommandProcessor`	Command map, Strategy (solver, scoring)	Atomic rating update; don't run a contest twice	Ch 21
PropertyHunt (ClearTrip)	`Property`, `SearchCriteria` (+ Builder), `PropertyService`	Builder + predicate list, Strategy (sort)	Synchronized mark-sold	Ch 21
Ledger Co (Navi)	`Loan`, `LumpSum`, `Balance`, `InterestPolicy`, `LedgerService`	Strategy (interest), derived state, command loop	Synchronized lump sums	Ch 21
Inventory management	`Product`, `Warehouse`, `StockMovement`, `AllocationStrategy`, `ReorderListener`	Strategy (allocation), Observer (edge-triggered alerts)	Overselling → lock per SKU	Ch 21 · Inventory
Unix file search (`find`)	`Node` → `FileNode` / `DirectoryNode`, `FileFilter`, `AndFilter` / `OrFilter` / `NotFilter`, `SearchCriteria` (+ Builder), `FileSearcher`, `FindCommandParser`	Specification + Composite (filters and file tree), Builder, recursive-descent parser	Weakly consistent children map; read-only fork/join walk	Ch 21 · Unix File Search

> **TL;DR — what every case study teaches**

- **Name the real unit of inventory** (copy, show seat, room-night, vehicle-day) (Ch 15 summary).
- **Every "if free then take" is one atomic step** — a monitor, a CAS, or a conditional `UPDATE` (Ch 15 summary).
- **Put rules on the object that owns the data**; the service only orders the steps (Ch 16, Ch 20 summaries).
- **Strategy** for the rule most likely to change; **Observer** for "tell someone when it happens"; **State** (or an enum state machine) for lifecycles.
- **Inject** time and randomness; money in `BigDecimal` or integer cents; call listeners **outside** locks.

22.10 Machine-coding and edge-case checklists

Source: Ch 21 · Machine-coding round (and Ch 1 · Interview code).

Machine-coding checklist

Before coding (0–20 min)

- [ ] Read the statement twice; mark **mandatory vs bonus**; ask clarifying questions.
- [ ] Put the statement's sample input into the driver as a `List<String>` of commands.

- [ ] Ask: "Where does it need to be thread-safe?" (wallets and inventory: always yes).
- [ ] Nouns → entities, verbs → service methods. Pick the 1–2 patterns the statement begs for ("can be sorted by X or Y" → Strategy).
- [ ] Create the six packages: **model, repository, service, strategy, exception, driver**.

While coding (20–75 min)

- [ ] Implement mandatory requirements **in the statement's order**; run the driver after each one (at least every 15 minutes).
- [ ] Entities guard their own invariants (e.g. `Task.changeStatus` rejects `TODO → DONE`).
- [ ] Services return data and throw domain exceptions; **no `System.out` in services**.
- [ ] Repository = interface + `ConcurrentHashMap` implementation; IDs from an `AtomicInteger`.
- [ ] Driver = `Map<String, Consumer<String[]>>` command processor; catch domain exceptions per command, **print and continue**.

Before you stop (75–90 min)

- [ ] Edge cases and validation done (see below); custom exceptions with readable messages.
- [ ] Thread-safety where money or stock is shared; say where the lock is and why.
- [ ] Bonus only after every mandatory feature runs.
- [ ] Clean names, delete dead code, run everything once more, prepare a 2-minute walkthrough.

Rubric the interviewer uses (Ch 21 · Machine-coding round)

(1) **Working, demo-able code** — thin end-to-end path in the first hour, never finish non-compiling; (2) **functional completeness** — mandatory first, in order; (3) **clean models** — enums for fixed sets, records for values, classes for changing things; (4) **separation of concerns** — Driver → Service → Repository; (5) **extensibility** — new rule = new class; (6) **edge cases** — custom exceptions, the driver keeps going; (7) **readability** — names from the statement's nouns and verbs; (8) **concurrency (when asked)** — say where the lock is, demo with a pool; (9) **tests (bonus)** — an `assert`-style self-check in `main` is enough.

Edge-case checklist (Ch 1 · Interview code, Ch 21 · Machine-coding round, case studies)

Category	Questions to ask your code
Input	`null`, empty, negative, zero, duplicate IDs, unknown IDs, invalid numbers
Ownership	Wrong user acting on someone else's object; a player moving out of turn
Capacity	Full lot, zero seats left, empty playlist, max loans reached, party larger than any table
State	Paying twice, exiting without entering, cancelling a delivered order, moving after game over, running a contest twice
Time	Exit before entry, holds/bookings expiring, part hours (round up?), an event exactly at the end instant (half-open intervals), late returns
Money	`BigDecimal` scale and rounding; no lost cents in splits; compare with `compareTo`, not `equals` (Ch 4 money note)
Concurrency	Two threads booking the same seat/spot/rider; multi-item bookings locking in sorted order; retries that must be idempotent
Failure	Payment declined after a hold; hold lost before payment → compensating refund; listener throws
Randomness	Seeded `Random` or a scripted strategy so the demo is repeatable

Common mistakes (Ch 21 · Machine-coding round)

Besides the Chapter 1 mistakes in 22.2: 20 minutes on a perfect diagram (enums/records first); one `Manager` with 15 maps (a service per area + repositories); `System.out` in services (the driver prints); `double` money (`BigDecimal` or whole units in a `long`); crashing on a bad command (catch per command); bonus before mandatory; unseeded `Random` (seed it or use a Strategy); `LocalDate.now()` in logic (inject a `Clock`).

---|---| | 20 minutes on a perfect class diagram | Enums/records first; the diagram lives in names | | One `Manager` with 15 maps | One service per area + repositories | | `System.out.println` inside a service | Services return data; the driver prints | | `double` for money | `BigDecimal` with a fixed scale, or whole units in a `long` | | Crashing on the first bad command | Catch domain exceptions in the driver loop | | Bonus before mandatory | Mandatory in order; bonus last | | `Random` without a seed | Seed it, or put randomness behind a Strategy | | `LocalDate.now()` in logic | Inject a `java.time.Clock` |

22.11 Glossary

Alphabetical. "Where" gives the chapter and topic where the term is explained in full. **Not repeated here** (see the tables above): the 23 GoF patterns and the extras (Simple Factory, Object Pool, Null Object, Dependency Injection, Repository, Specification, MVC, Event Sourcing) → 22.7; the five SOLID principles → 22.4; DRY, KISS, YAGNI, Law of Demeter, Tell Don't Ask, Program to an interface, Encapsulate what varies, Loose coupling, Hollywood Principle, Separation of Concerns, Least Astonishment, Fail fast, Design by Contract and the nine GRASP guidelines → 22.5; the 24 code smells → 22.6; the four pillars, overloading/overriding, interfaces vs abstract classes, the six relationships and the other one-line OOP facts → 22.3; the `java.util.concurrent` primitives, Coffman conditions, deadlock, livelock and starvation → 22.8.

Term	Plain meaning	Where
ABA problem	A value changes A → B → A, so a compare-and-swap cannot notice the change; fix with a version stamp (`AtomicStampedReference`)	Ch 12 · CAS & atomics
Activity diagram	UML flowchart of a process, with decisions, swimlanes and fork/join	Ch 3 · Activity diagrams
Actor	Someone or something that uses the system (customer, admin, the system itself)	Ch 1 · Requirements
Aggregate (DDD)	A cluster of objects changed as one unit; only its root is reachable from outside, and the root guards the invariants	Ch 5 · DDD
Anemic domain model	Classes with data but no behaviour; the logic lives in services that pull the data out	Ch 5 · Tell, Don't Ask
Anti-corruption layer	Translation layer at a bounded context's border that keeps a foreign or legacy model out of yours	Ch 5 · DDD
Anti-pattern	A common "solution" that reliably causes more problems than it solves	Ch 8 · Anti-patterns
Append-only log	Messages are only ever added at the end; readers track their own offset	Ch 20 · Pub-sub
Association class	A class that holds data about a link itself (e.g. `Enrollment{grade}`)	Ch 3 · Class diagrams
At-least-once delivery	A message may be delivered more than once, never lost; consumers must be idempotent	Ch 20 · Pub-sub
Atomic variable	`AtomicInteger` / `AtomicLong` / `AtomicReference`: lock-free single-variable updates using CAS	Ch 12 · CAS & atomics
Back-pressure	Slowing producers when consumers can't keep up, e.g. with a bounded queue	Ch 12 · Producer-consumer
`BigDecimal`	Exact decimal type used for money; build from strings, compare with `compareTo`	Ch 1 · Interview code, Ch 4 (money note)
Bloom filter	Probabilistic set: may say "maybe present" wrongly, never says "absent" wrongly	Ch 13 · Concurrent Bloom Filter
Bounded context	Boundary inside which one domain model and one vocabulary apply consistently (often one team or service)	Ch 5 · DDD
Boy Scout Rule	Leave the code a little cleaner than you found it	Ch 6 (intro)
Caretaker / Originator	Memento roles: the originator makes and restores snapshots; the caretaker stores them without looking inside	Ch 10 · Memento
CAS (compare-and-swap)	Atomic "set to new value only if it still equals the expected value"	Ch 12 · CAS & atomics
Characterisation test	A test that records what existing code does now, before you refactor it	Ch 7 · Full source code and output
Check-then-act	Read a condition, then act on it, as two steps: a race unless made atomic	Ch 12 · Race conditions, Concurrency in LLD interviews
Circular wait	T1 waits for T2's lock while T2 waits for T1's; one of the four deadlock conditions	Ch 12 · Deadlock
Class	A blueprint for objects: fields plus methods	Ch 2 · Classes & keywords
Class diagram	UML diagram of classes, fields, methods and relationships — "the answer" in an LLD round	Ch 3 · Class diagrams

Term	Plain meaning	Where
Class explosion	Subclasses multiplying across two dimensions (`WindowsButton` , `MacButton` , …)	Ch 9 · Bridge Pattern
Clean code	Code whose intent is obvious to the next reader and which is easy to change safely	Ch 6 (intro)
Code review	A peer reads a change before merge: correctness first, then design, readability, tests, security, performance	Ch 6 · Code review checklist
Code smell	A surface symptom that usually signals a deeper design problem; not a bug by itself	Ch 7 · Code smells
Command–Query Separation (CQS)	A method either changes state or returns a value, not both	Ch 6 · Functions
Compensation	Undoing a completed step with a new action (e.g. a refund) because money can't be rolled back	Ch 15 · BookMyShow
Composition root	The one place (usually `main`) where objects are created and wired together	Ch 6 · Systems
Concurrency vs parallelism	Many tasks in progress (structure) vs many executing at the same instant (hardware)	Ch 12 · Concurrency vs parallelism
Condition variable	Lets a thread wait until shared state meets a condition (`wait/notify` , `Condition`)	Ch 12 · Condition variables
Constructor chaining	One constructor calling another with `this(...)`	Ch 2 · Classes & keywords
Critical section	Code that touches shared state and must run in one thread at a time	Ch 12 · Race conditions
Dead-letter queue	Where messages go after retries are exhausted, for later inspection	Ch 20 · Pub-sub
Deep vs shallow copy	Copying nested objects too vs sharing them with the original	Ch 8 · Prototype
Derive, don't store	Compute counts and totals from the underlying records so they can't drift	Ch 18 · summary
Design pattern	A named, reusable solution to a recurring design problem	Ch 8 · What is a design pattern?
Director	Optional Builder role that runs a fixed sequence of building steps	Ch 8 · Builder
Domain event	Past-tense record of a business fact (`OrderPlaced`) that other parts of the system react to	Ch 5 · DDD
Domain-Driven Design (DDD)	Modelling software on the business: bounded contexts, entities, value objects, aggregates, domain events, repositories	Ch 5 · DDD
Double dispatch	Choosing a method by both the element's and the visitor's type (Visitor's `accept`)	Ch 11 · Visitor
Double-checked locking	Lazy singleton that checks, locks, checks again; the field must be `volatile`	Ch 8 · Singleton
Dynamic dispatch	The JVM picks an overridden method from the object's actual class at run time	Ch 2 · Polymorphism, Compile-time vs runtime polymorphism
Entity (DDD)	Object defined by an identity that survives changes; two entities are equal when their ids are equal	Ch 5 · DDD
Expression problem	Visitor makes adding operations cheap but adding types expensive (and the reverse for plain polymorphism)	Ch 11 · Visitor

Term	Plain meaning	Where
Fairness	Serving waiting threads roughly in order so none starves; costs throughput	Ch 12 · Starvation
F.I.R.S.T.	Clean tests are Fast, Independent, Repeatable, Self-validating, Timely	Ch 6 · Unit tests
Flag argument	A boolean parameter that makes a function do two things; split it instead	Ch 6 · Functions
Fork/Join	Split a task recursively into subtasks run in parallel, then combine results	Ch 12 · Choosing the tool, Ch 13 · Parallel merge sort
Fragile base class	A change in a parent class silently breaks subclasses	Ch 5 · Composition over inheritance
Functional interface	An interface with one abstract method, usable with lambdas	Ch 2 · Interfaces
Functional requirement	A behaviour the system must have ("a customer takes a ticket")	Ch 1 · Requirements
Glob	Filename wildcard pattern: `*` any characters, `?` one character, `[0-9]` one character from a set	Ch 21 · Unix File Search
God class / God Object	One class that knows and does almost everything	Ch 8 · Anti-patterns, Ch 7 · Code smells
Golden Hammer	Applying a favourite tool or pattern to every problem	Ch 8 · Anti-patterns
GRASP	Larman's nine guidelines for assigning responsibilities to classes	Ch 5 · GRASP
Guard clause	An early check-and-return/throw at the top of a method instead of nested `if`s	Ch 7 · Refactoring catalogue, Ch 5 · Fail fast
Happens-before	Java Memory Model rule guaranteeing one action's effects are visible to another	Ch 12 · Java Memory Model
HLD vs LLD	Many services and machines vs classes inside one application	Ch 1 · The interview
Hold with TTL	A temporary claim (e.g. seats during payment) that expires automatically	Ch 15 · BookMyShow
Idempotency	Running an operation twice has the same effect as once; an idempotency key enforces it	Ch 21 · Digital wallet, Ch 20 · Pub-sub
Immutability	Objects that never change after creation; safe to share between threads	Ch 2 · Immutability & records, Ch 5 · Immutability
Intrinsic vs extrinsic state	Flyweight's shared, immutable part vs the per-use part passed in by the caller	Ch 9 · Flyweight Pattern
Invariant	A fact about an object that must always be true ("balance is never negative")	Ch 5 · Design by Contract
Inversion of Control (IoC)	The framework controls the flow and calls your code; DI is one way to achieve it	Ch 5 · Hollywood Principle, Ch 4 · SOLID Q&A
Inverted index	A map from each word/tag to the list of items containing it, for search	Ch 18 · Stack Overflow
Invoker / Receiver	Command roles: the invoker triggers commands; the receiver does the real work	Ch 10 · Command
Java Memory Model (JMM)	Rules for when one thread's writes become visible to another	Ch 12 · Java Memory Model

Term	Plain meaning	Where
Lock ordering	Always acquire locks in one global order to prevent circular wait	Ch 12 · Deadlock
Lock striping	Guarding a map with an array of locks, each covering some buckets	Ch 13 · Striped concurrent map
LOOK scheduling	Elevator keeps moving in one direction while stops remain ahead, then reverses	Ch 14 · Elevator System
Lost wakeup	A signal sent before the waiter started waiting, so the waiter sleeps forever	Ch 12 · Condition variables
LRU cache	Evicts the least recently used entry; hash map + doubly linked list gives O(1)	Ch 20 · Cache / LRU
Machine-coding round	LLD round where working, runnable in-memory code is the answer (90–120 min)	Ch 1 · The interview, Ch 21 · Machine-coding round
Matching engine	Per-symbol order book matching buy and sell orders by best price, then oldest first	Ch 19 · Stock Brokerage
Multiplicity	How many objects take part at each end of a relationship (`1`, `0..1`, `*`, `1..*`)	Ch 3 · Association
Mutex	A lock that lets only one thread into a critical section	Ch 12 · Mutex and synchronized
Non-functional requirement	A quality: thread safety, extensibility, in-memory storage, latency	Ch 1 · Requirements
Object	One instance of a class with real values	Ch 2 · Classes & keywords
Optimistic locking	Read with a version; the write succeeds only if the version is unchanged	Ch 12 · Read-write locks, Concurrency in LLD interviews
`Optional`	A container that may be empty; return it when "not found" is normal	Ch 1 · Interview code
Out-of-scope list	Written list of what you will not build; shrinks the problem	Ch 1 · Requirements
Pinning	A virtual thread stuck on its carrier thread while it blocks	Ch 12 · Creating threads
PLU code	Four-digit price look-up code for loose produce, keyed at the till together with the scale weight	Ch 19 · Grocery Store
Poison pill	A special message telling a consumer to stop; one per consumer	Ch 12 · Producer-consumer
Producer–consumer	Threads hand work through a (bounded) queue	Ch 12 · Producer-consumer
Pub-sub	Observer through a broker: publishers and subscribers don't know each other	Ch 20 · Pub-sub
Qualified association	An association looked up by a key (`Library [isbn] → Book`, i.e. a `Map`)	Ch 3 · Class diagrams
Race condition	The result depends on the timing of threads touching shared mutable state	Ch 12 · Race conditions
Rate limiter	Caps how many requests a key may make per time window (token bucket, windows)	Ch 13 · Rate limiter, Ch 20 · Rate Limiter
Record	Compact Java class for immutable data; only shallowly immutable	Ch 2 · Immutability & records
Reentrancy	A thread may re-acquire a lock it already holds	Ch 12 · ReentrantLock and reentrancy

Term	Plain meaning	Where
Refactoring	Changing internal structure without changing observable behaviour	Ch 7 · Refactoring fundamentals
Requirement	One thing the system must do, written so you can check it	Ch 1 · Requirements
Role interface	A small interface serving one kind of caller (the ISP fix)	Ch 4 · Interface Segregation Principle
Rule of three	Refactor to an abstraction at the second or third case, not the first	Ch 4 · Over-engineering
Sealed type	A type whose permitted subtypes are fixed, enabling exhaustive `switch`	Ch 11 · Visitor
Sequence diagram	UML diagram of messages between objects over time	Ch 3 · Sequence diagrams
Service Locator	Anti-pattern where a class asks a registry for its dependencies, hiding them	Ch 11 · Beyond GoF
Sharding	Splitting data across independent parts (each with its own lock) to scale	Ch 20 · Cache / LRU
Signaling	One thread tells another that something has happened (semaphore, latch)	Ch 12 · Signaling pattern
Sliding window	Rate-limit algorithm counting requests in a moving time window (log or counter)	Ch 20 · Rate Limiter
SOLID	SRP, OCP, LSP, ISP, DIP: five guidelines that keep one change from spreading	Ch 4 · What SOLID is
Split tender	Paying one bill with several payment methods (e.g. part UPI, part cash)	Ch 19 · Grocery Store
Spurious wakeup	A waiting thread wakes with no signal; why `wait()` sits in a `while` loop	Ch 12 · Condition variables
State machine diagram	UML diagram of one object's states and transitions (`event [guard] / action`)	Ch 3 · State machine diagrams
`static`	Belongs to the class, one per class, not to each object	Ch 2 · Classes & keywords
Test double	Anything standing in for a real dependency in a test: dummy, stub, spy, mock, fake	Ch 6 · Testing strategies
Test-Driven Development (TDD)	Failing test → make it pass → refactor, in very short cycles	Ch 6 · Unit tests
Thread	An independent path of execution inside a process, sharing its memory	Ch 12 · Processes vs threads
Thread confinement	Keeping data inside one thread so it never needs locking	Ch 12 · Immutability & confinement
Thread pool	A fixed set of reusable threads fed from a task queue	Ch 12 · Thread pool pattern
Thread states	NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED	Ch 12 · Thread states
Thread-starvation deadlock	Tasks in a bounded pool wait for tasks queued behind them in the same pool	Ch 12 · Deadlock, Thread pool pattern
Token bucket	Rate-limit algorithm: tokens refill at a fixed rate; each request takes one; allows bursts	Ch 13 · Rate limiter
Two hats	While coding you are either adding function or refactoring, never both at once	Ch 7 · Refactoring fundamentals

Term	Plain meaning	Where
Ubiquitous language	One vocabulary shared by domain experts and the code (`place()` , not `setStatus(2)`)	Ch 5 · DDD
UML	Unified Modeling Language: 14 diagram types, 7 structural and 7 behavioural	Ch 3 · What UML is
Use case	One thing an actor can do with the system	Ch 1 · Requirements, Ch 3 · Use-case diagrams
Use-case diagram	UML diagram of actors, use cases and the system boundary («include», «extend»)	Ch 3 · Use-case diagrams
Value object	A small immutable object defined by its values (money, date range), usually a record	Ch 2 · Immutability & records, Ch 5 · DRY, Ch 5 · DDD
Virtual waiting room	FIFO queue that admits buyers gradually during a ticket rush	Ch 15 · Concert tickets

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Self-check

- **Q:** Name the eleven framework steps in order. **A:** Clarify, requirements, use cases, entities, relationships, class diagram, key flows, patterns & SOLID, code, edge cases & concurrency, extensions.
- **Q:** Where does the diamond go, and what does filled mean? **A:** At the whole; filled = composition (the part dies with the whole).
- **Q:** A `switch (vehicleType)` computes fees. Which principle, smell and pattern? **A:** OCP; Repeated Switches (if it repeats); Strategy.
- **Q:** Decorator vs Proxy vs Chain of Responsibility? **A:** Decorator adds behaviour and every layer runs; Proxy controls access; in CoR one handler can stop the flow.
- **Q:** State vs Strategy? **A:** States know each other and advance themselves; strategies are chosen from outside.
- **Q:** Four Coffman conditions, and the usual fix? **A:** Mutual exclusion, hold and wait, no preemption, circular wait; global lock ordering.
- **Q:** Why is `volatile` not enough for `count++`? **A:** Visibility, not atomicity; use `AtomicInteger` or a lock.
- **Q:** The six machine-coding packages? **A:** model, repository, service, strategy, exception, driver.
- **Q:** What makes every booking problem safe? **A:** Naming the real unit of inventory and making "if free then take" one atomic step (plus holds that expire).

TL;DR

- **Framework:** clarify → requirements → use cases → entities → relationships → diagram → flows → patterns/SOLID → code → edge cases/concurrency → extensions.
- **Classes:** different data → field; different behaviour → class; one sentence per class; composition over inheritance.
- **SOLID** diagnoses change pain; always state the cost of the fix, and know when not to apply it.
- **Patterns** come from signals: Strategy (rules vary), Factory (type value → class), Observer (many listeners), State (lifecycle), Singleton (exactly one), Builder (many optional fields). Use one or two, name a rejected one.
- **Concurrency:** find the shared state, state the invariant, make check-then-act atomic with the smallest tool, lock in a fixed order, notify outside locks.
- **Machine coding:** run early and often, mandatory first, services return data, the driver prints and keeps going.